Midterm—Algebra
Fall 2023

Algebra Team (2025)

1. A matrix which is obtained by performing six elementary row operations on I_n is:

   (a) nonsingular (b) elementary (c) singular
   (d) rectangular (e) none

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2. A, B and C are $n \times n$ matrices. A is diagonal, B is lower triangular and C is upper triangular. $(C + A)B^T$ is:

   (a) lower triangular (b) symmetric (c) upper triangular
   (d) diagonal (e) none

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3. If A is an $n \times m$ matrix and B is an $n \times n$ skew-symmetric matrix, then $A^T BA$ is:

   (a) symmetric (b) skew-symmetric (c) diagonal
   (d) triangular (e) none

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4. If $x = -2$ is a root of the equation $2x^5 + 5x^4 + a_3x^3 - 23x^2 + 79x - 30 = 0$, then a_3 is equal to:

   (a) -33 (b) 33 (c) 16.5 (d) -16.5 (e) none

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5. If the remainder of dividing $P(x)$ by $(x^2 + 3x)$ is equal to $(3x + 5)$, then $P(-3)$ is equal to:

   (a) -2 (b) -3 (c) -5 (d) -4 (e) none

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6. If $A = \begin{bmatrix} 1 & 3 & -7 \\ -2 & 1 & r \end{bmatrix}$, $r \in \mathbb{R}$, then the entry in row 1 and column 2 of AA^T is equal to:

   (a) $7r - 1$ (b) $-7 - 2r$ (c) $-7r + 1$ (d) $7 + 2r$ (e) none

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7. The linear system $\begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix} \mathbf{x} = \begin{bmatrix} -1 \\ 0 \\ 2 \\ w \end{bmatrix}$ will have a unique solution if w is equal to:

   (a) -8 (b) -2 (c) -4 (d) -6 (e) none

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8. If $P(x) = x^3 + 4x^2 - 8x + a_0$ is divided by $(x - (2 - i))$, then the remainder is:

   (a) $a_0 - 4 - 18i$ (b) $a_0 - 2 - 19i$ (c) $a_0 + 2 - 21i$ (d) $a_0 + 4 - 22i$ (e) none

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9. If $(A^T + 3I)^{-1} = \begin{bmatrix} 1 & 4 \\ 0 & g \end{bmatrix}$, $g \neq 0$, then the entry in row 2 and column 1 of A is equal to:

   (a) $-4/g$ (b) $4/g$ (c) $-1/g$ (d) $1/g$ (e) none

10. The linear system $\begin{cases} x - 4y + 2z = 0 \\ x + 3y - 4z = 0 \\ rx + sy + tz = 0 \end{cases}$ will have nontrivial solutions if:

(a) $2r + s + t = 0$ (b) $r - 2s + 3t = 0$ (c) $2r + 5s + 2t = 0$
(d) $10r + 6s + 7t = 0$ (e) none

$x = -1$ is a repeated root of multiplicity 2 of the polynomial equation
$3x^5 + 2x^4 - 22x^3 - 32x^2 + a_1x + a_0 = 0$

11. The value of a_1 is:

(a) 5 (b) -5 (c) 1.67 (d) -1.67 (e) none

12. The value of a_0 is:

(a) -6 (b) 2 (c) 6 (d) -2 (e) none

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13. Consider the linear system $\begin{cases} x + 4y + 3z = 2 \\ x + 5y + z = -4 \\ 3x + 2y + 29z = 66 \end{cases}$, if $z = t$, $t \in \mathbb{R}$ then x is equal to:

(a) $26 + 11t$ (b) $13 + 4t$ (c) $13 - 4t$ (d) $26 - 11t$ (e) none

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14. The linear system $\begin{cases} x - 2y + 4z = a \\ 2x - 3y + 11z = b \\ 3x - 3y + 21z = c \end{cases}$ will be consistent if:

(a) $c - 3a - 3b$ (b) $c = -5a + 4b$ (c) $c = 5a - 4b$
(d) $c = -3a + 3b$ (e) none

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15. Consider the equation $x^3 + x^2 + ax + b = 0$, where a and b are nonzero real numbers. The equation will have one positive real root if:

(a) b is positive (b) a and b have the same sign (c) b is negative
(d) a and b have opposite signs (e) none

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16. The entry in row 1 and column 7 of the reduced row echelon form of the matrix $A = \begin{bmatrix} 1 & 2 & -1 & 4 & 1 & 1 & 3 \\ 2 & 5 & 0 & 9 & 2 & 4 & 8 \\ 3 & 8 & 1 & 15 & 2 & 8 & 14 \end{bmatrix}$
is equal to:

(a) -3 (b) -4 (c) -2 (d) -1 (e) none

Consider the LU decomposition of the matrix $\begin{bmatrix} 2 & 2 & 4 \\ 3 & 2 & 11 \\ 4 & 5 & f \end{bmatrix}$.

17. The entry in row 2 and column 3 of U is equal to:

(a) -11 (b) -5 (c) -6 (d) -4 (e) none

18. The entry in row 3 and column 3 of L is equal to:

(a) $1/(f-3)$ (b) $f-3$ (c) $f-4$ (d) $1/(f-4)$ (e) none

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19. Consider the LU decomposition of the matrix $\begin{bmatrix} 2 & 1 & 2 & 0 \\ 2 & 2 & 3 & 2 \\ 4 & 3 & 6 & 3 \\ 0 & 2 & 3 & -3 \end{bmatrix}$. The entry in row 4 and column 4 of L is equal to:

(a) -8 (b) $-1/8$ (c) -7 (d) $-1/7$ (e) none

Consider the linear system $\begin{cases} x+2y+2z=1 \\ 2x+3y+9z=-1 \\ 3x+4y+kz=1 \end{cases}$, $k \neq 16$.

20. The value of y is equal to:

(a) $(3k-25)/(k-16)$ (b) $(4k-18)/(k-16)$ (c) $(3k-28)/(k-16)$
(d) $(4k-15)/(k-16)$ (e) none

21. The value of x is equal to:

(a) $(-5k+32)/(k-16)$ (b) $(-5k+37)/(k-16)$ (c) $(-3k+22)/(k-16)$
(d) $(-3k+25)/(k-16)$ (e) none

The matrix $A = \begin{bmatrix} 1 & 2 & 6 & 3 \\ 2 & 5 & 15 & 11 \\ 3 & 2 & 7 & -7 \\ -2 & -1 & -1 & k \end{bmatrix}$, where $k \neq 17$, is nonsingular.

22. The entry in row 1 and column 4 of A^{-1} is:

(a) $9/(k-17)$ (b) $7/(k-17)$ (c) $5/(k-17)$
(d) $3/(k-17)$ (e) none

23. The entry in row 1 and column 1 of A^{-1} is:

(a) $5(k+34)/(k-17)$ (b) $5(k+16)/(k-17)$ (c) $5(k+7)/(k-17)$
(d) $5(k+25)/(k-17)$ (e) none

Midterm—Algebra
Fall 2023

Algebra Team (2025)

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1. A matrix which is obtained by performing six elementary row operations on I_n is:

Answer: (a) nonsingular; that means

$$E_6 E_5 \dots E_1 I_n = B$$

which means, (if we multiply by the inverses of the elementary matrices)

$$\underbrace{E_1^{-1} E_2^{-1} \dots E_6^{-1}}_{B^{-1}} B = I_n$$

so that B is nonsingular.

2. A, B and C are $n \times n$ matrices. A is diagonal, B is lower triangular and C is upper triangular. $(C + A)B^T$ is:

Answer: (c) upper triangular; since A is a diagonal matrix and C is an upper triangular matrix, then their sum gives an upper triangular matrix

$$\begin{bmatrix} a_1 & 0 & \dots & 0 \\ 0 & a_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_n \end{bmatrix} + \begin{bmatrix} c_{11} & c_{21} & \dots & c_{n1} \\ 0 & c_{22} & \dots & c_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & c_{nn} \end{bmatrix} = \begin{bmatrix} a_1 + c_{11} & c_{21} & \dots & c_{n1} \\ 0 & a_2 + c_{22} & \dots & c_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_n + c_{nn} \end{bmatrix}$$

and the transpose of a lower triangular matrix is an upper triangular matrix.

Finally, the multiplication of two upper triangular matrices yields an upper triangular matrix.

3. If A is an $n \times m$ matrix and B is an $n \times n$ skew-symmetric matrix, then $A^T BA$ is:

Answer: (b) skew-symmetric; let D be an $n \times n$ matrix, then

(a) if D is skew-symmetric; $D^T = -D$

(b) if D is symmetric; $D^T = D$

then, taking the transpose of A^TBA

$$\begin{aligned}(A^TBA)^T &= A^TB^TA \\ &= -A^TBA \qquad\qquad (B^T = -B)\end{aligned}$$

and the resulting matrix is of size $m \times m$.

4. If $x = -2$ is a root of the equation $2x^5 + 5x^4 + a_3x^3 - 23x^2 + 79x - 30 = 0$, then a_3 is equal to:

Answer: (a) -33; if $x = -2$ is a root of the polynomial equation, then $P(-2) = 0$. Substituting

$$2(-2)^5 + 5(-2)^4 + a_3(-2)^3 - 23(-2)^2 + 79(-2) - 30 = 0$$

$$-8a_3 - 264 = 0 \Rightarrow a_3 = -33$$

5. If the remainder of dividing $P(x)$ by $(x^2 + 3x)$ is equal to $(3x + 5)$, then $P(-3)$ is equal to:

Answer: (d) -4; if $P(x)$ is divided by $(x^2 + 3x)$ and has a remainder of $(3x + 5)$, then

$$P(x) = (x^2 + 3x)Q(x) + 3x + 5$$

then substituting $x = -3$

$$P(-3) = \underbrace{((-3)^2 + 3 \times -3)Q(x)}_{=0} + 3 \times -3 + 5 = -4$$

6. If $A = \begin{bmatrix} 1 & 3 & -7 \\ -2 & 1 & r \end{bmatrix}$, $r \in \mathbb{R}$, then the entry in row 1 and column 2 of AA^T is equal to:

Answer: (c) $-7r + 1$; firstly, A^T

$$A^T = \begin{bmatrix} 1 & -2 \\ 3 & 1 \\ -7 & r \end{bmatrix}$$

then AA^T can be carried out readily

$$\begin{bmatrix} 1 & 3 & -7 \\ -2 & 1 & r \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 3 & 1 \\ -7 & r \end{bmatrix}$$

paying close attention to the requirement, only the entry in row 1 and column 2 is required

$$1 \times -2 + 3 \times 1 - 7 \times r = -7r + 1$$

7. The linear system $\begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix} \mathbf{x} = \begin{bmatrix} -1 \\ 0 \\ 2 \\ w \end{bmatrix}$ will have a unique solution if w is equal to:

Answer: (d) -6; putting it into augmented matrix form

$$\left[\begin{array}{ccc|c} 1 & 2 & 4 & -1 \\ 0 & 1 & 5 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 1 & 2 & w \end{array}\right]$$

proceeding with Gaussian elimination

$$\xrightarrow{r_4 - r_2 \to r_4} \left[\begin{array}{ccc|c} 1 & 2 & 4 & -1 \\ 0 & 1 & 5 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & -3 & w \end{array}\right]$$

$$\xrightarrow{r_4 + 3r_3 \to r_4} \left[\begin{array}{ccc|c} 1 & 2 & 4 & -1 \\ 0 & 1 & 5 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & w+6 \end{array}\right]$$

for the system to be consistent

$$\#\text{leading variables} = \#\text{non-zero rows} = \#\text{unknowns} = 3$$

then $w + 6 = 0 \Rightarrow w = -6$.

8. If $P(x) = x^3 + 4x^2 - 8x + a_0$ is divided by $(x - (2 - i))$, then the remainder is:

Answer: (b) $a_0 - 2 - 19i$; proceeding with synthetic division $(x = 2 - i)$

$$\begin{array}{c|ccc|c} 2-i & 1 & 4 & -8 & a_0 \\ + & & 2-i & 11-8i & -2-19i \\ \hline & 1 & 6-i & 3-8i & a_0 - 2 - 19i \end{array}$$

then the remainder is clearly $a_0 - 2 - 19i$.

9. If $(A^T + 3I)^{-1} = \begin{bmatrix} 1 & 4 \\ 0 & g \end{bmatrix}$, $g \neq 0$, then the entry in row 2 and column 1 of A is equal to:

Answer: (a) $-4/g$; taking the inverse of both sides then adding $-3I$ to both sides

$$A^T = \frac{1}{g} \begin{bmatrix} g & -4 \\ 0 & 1 \end{bmatrix} - 3I$$

taking the transpose of both sides

$$A = \frac{1}{g} \begin{bmatrix} g & 0 \\ -4 & 1 \end{bmatrix} - 3I$$

then the entry in row 2 and column 1 of A is $-4/g$.

10. The linear system $\begin{cases} x - 4y + 2z = 0 \\ x + 3y - 4z = 0 \\ rx + sy + tz = 0 \end{cases}$ will have nontrivial solutions if:

Answer: (d) $10r + 6s + 7t = 0$; generally, a homogeneous linear system

$$A\mathbf{x} = \mathbf{0}$$

will only have the unique solution (the trivial solution), if

$$\mathbf{x} = A^{-1}\mathbf{0} \qquad \text{(i.e. } A \text{ is invertible} \Rightarrow |A| \neq 0)$$

but if $|A| = 0$, then there are infinitely many solutions (which include the trivial solution **AND** the nontrivial solutions).

Then, calculating the determinant of A using the final row

$$\begin{vmatrix} 1 & -4 & 2 \\ 1 & 3 & -4 \\ r & s & t \end{vmatrix} = r \times (16 - 6) - s \times (-4 - 2) + t \times (3 + 4) = 10r + 6s + 7t$$

and since the determinant has to equal zero, then $10r + 6s + 7t = 0$.

$x = -1$ is a repeated root of multiplicity 2 of the polynomial equation $3x^5 + 2x^4 - 22x^3 - 32x^2 + a_1x + a_0 = 0$

11. The value of a_1 is:

Answer: (b) -5; since $x = -1$ is a repeated root of multiplicity 2, then if we do synthetic division by $x = -1$ twice, the remainder (in both cases) should be zero. Doing synthetic division once

-1	3	2	-22	-32	a_1	a_0
$+$		-3	1	21	11	$-11 - a_1$
	3	-1	-21	-11	$11 + a_1$	$-11 + a_0 - a_1$

and so, $a_0 - a_1 = 11$. Doing synthetic division again (on the quotient)

-1	3	-1	-21	-11	$11 + a_1$
$+$		-3	4	17	-6
	3	-4	-17	6	$5 + a_1$

then $5 + a_1 = 0 \Rightarrow a_1 = -5$

12. The value of a_0 is:

Answer: (c) 6; from the previous problem

$$a_0 - a_1 = 11 \Rightarrow a_0 = 6$$

13. Consider the linear system $\begin{cases} x+4y+3z=2 \\ x+5y+z=-4 \\ 3x+2y+29z=66 \end{cases}$, if $z=t$, $t\in\mathbb{R}$ then x is equal to:

Answer: **(d)** $26-11t$; putting the system into augmented matrix form

$$\left[\begin{array}{ccc|c} 1 & 4 & 3 & 2 \\ 1 & 5 & 1 & -4 \\ 3 & 2 & 29 & 66 \end{array}\right]$$

proceeding with Gaussian elimination

$$\xrightarrow[r_3-3r_1\to r_3]{r_2-r_1\to r_2} \left[\begin{array}{ccc|c} 1 & 4 & 3 & 2 \\ 0 & 1 & -2 & -6 \\ 0 & -10 & 20 & 60 \end{array}\right]$$

$$\xrightarrow{r_3+10r_2\to r_3} \left[\begin{array}{ccc|c} 1 & 4 & 3 & 2 \\ 0 & 1 & -2 & -6 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

then, by back-substitution, $y=-6+2t$ and $x=26-11t$.

14. The linear system $\begin{cases} x-2y+4z=a \\ 2x-3y+11z=b \\ 3x-3y+21z=c \end{cases}$ will be consistent if:

Answer: **(d)** $c=-3a+3b$; putting our system in augmented matrix form

$$\left[\begin{array}{ccc|c} 1 & -2 & 4 & a \\ 2 & -3 & 11 & b \\ 3 & -3 & 21 & c \end{array}\right]$$

then proceeding with Gaussian elimination

$$\xrightarrow[r_3-3r_1\to r_3]{r_2-2r_1\to r_2} \left[\begin{array}{ccc|c} 1 & -2 & 4 & a \\ 0 & 1 & 3 & b-2a \\ 0 & 3 & 9 & c-3a \end{array}\right]$$

$$\xrightarrow{r_3-3r_2\to r_3} \left[\begin{array}{ccc|c} 1 & -2 & 4 & a \\ 0 & 1 & 3 & b-2a \\ 0 & 0 & 0 & c+3a-3b \end{array}\right]$$

then it's obvious the system will be consistent if

$$c+3a-3b=0 \Rightarrow c=-3a+3b$$

15. Consider the equation $x^3 + x^2 + ax + b = 0$, where a and b are nonzero real numbers. The equation will have one positive real root if:

Answer: (c) b is negative; we'll start by recognizing that the words "... a and b are nonzero real numbers ..."mean that a and b can take on any value (negative or positive) except 0.

Then, start by considering answer (a) b is positive. Start with $b > 0$, $a = -\epsilon < 0$

$$P(x) = x^3 + x^2 + ax + b = x^3 + x^2 - \epsilon x + b \qquad \text{(changes signs twice, n}(P) \in \{2, 0\})$$

now let a, $b > 0$

$$P(x) = x^3 + x^2 + ax + b \qquad \text{(never changes signs, n}(P) = 0)$$

we see that answer gives either 2 or 0 positive roots. We require 1 positive root.

Consider answer (b) a and b have the same sign. Start with a, $b > 0$

$$P(x) = x^3 + x^2 + ax + b \qquad \text{(never changes signs, n}(P) = 0)$$

now let a, $b < 0$

$$P(x) = x^3 + x^2 \underbrace{+ax + b}_{\substack{\text{each} \\ < 0}} \qquad \text{(changes signs once, n}(P) = 1)$$

which gives either 1 or 0 positive roots. So it's not a guarantee that it gives one positive root.

Consider answer (d) a and b have opposite signs. Start with $a > 0$, $b = -\epsilon < 0$

$$P(x) = x^3 + x^2 + ax + b = x^3 + x^2 + ax - \epsilon \qquad \text{(changes signs once, n}(P) = 1)$$

now let $b > 0$, $a = -\epsilon < 0$

$$P(x) = x^3 + x^2 + ax + b = x^3 + x^2 - \epsilon x + b \qquad \text{(changes signs twice, n}(P) \in \{2, 0\})$$

which gives either 2, 0 or 1 positive root. We require a guaranteed 1 positive real root.

Finally consider (c) b is negative. Let $b = -\epsilon < 0$ and $a > 0$

$$P(x) = x^3 + x^2 + ax + b = x^3 + x^2 + ax - \epsilon \qquad \text{(changes signs once, n}(P) = 1)$$

now let both a, $b < 0$

$$P(x) = x^3 + x^2 \underbrace{+ax + b}_{\substack{\text{each} \\ < 0}} \qquad \text{(changes signs once, n}(P) = 1)$$

obviously, answer (c) gives a guaranteed 1 positive real root.

16. The entry in row 1 and column 7 of the reduced row echelon form of the matrix $A = \begin{bmatrix} 1 & 2 & -1 & 4 & 1 & 1 & 3 \\ 2 & 5 & 0 & 9 & 2 & 4 & 8 \\ 3 & 8 & 1 & 15 & 2 & 8 & 14 \end{bmatrix}$ is equal to:

Answer: (a) -3; performing row reduction

$$\xrightarrow[r_3 - 3r_1 \to r_3]{r_2 - 2r_1 \to r_2} \begin{bmatrix} 1 & 2 & -1 & 4 & 1 & 1 & 3 \\ 0 & 1 & 2 & 1 & 0 & 2 & 2 \\ 0 & 2 & 4 & 3 & -1 & 5 & 5 \end{bmatrix}$$

$$\xrightarrow[r_3 - 2r_2 \to r_3]{r_1 - 2r_2 \to r_1} \begin{bmatrix} 1 & 0 & -5 & 2 & 1 & -3 & -1 \\ 0 & 1 & 2 & 1 & 0 & 2 & 2 \\ 0 & 0 & 0 & 1 & -1 & 1 & 1 \end{bmatrix}$$

$$\xrightarrow[r_2 - r_3 \to r_2]{r_1 - 2r_3 \to r_1} \begin{bmatrix} 1 & 0 & -5 & 0 & 3 & -5 & -3 \\ 0 & 1 & 2 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & -1 & 1 & 1 \end{bmatrix}$$

so that the entry in row 1 and column 7 is -3.

Consider the LU decomposition of the matrix $\begin{bmatrix} 2 & 2 & 4 \\ 3 & 2 & 11 \\ 4 & 5 & f \end{bmatrix}$.

17. The entry in row 2 and column 3 of U is equal to:

Answer: (b) -5; using the careful book-keeping method

$$\xrightarrow[\to]{(1/2)r_1 \to r_1} \begin{bmatrix} 1 & 1 & 2 \\ 3 & 2 & 11 \\ 4 & 5 & f \end{bmatrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 \\ & & 0 \\ & & \end{bmatrix}$$

$$\xrightarrow[r_3 - 4r_1 \to r_3]{r_2 - 3r_1 \to r_2} \begin{bmatrix} 1 & 1 & 2 \\ 0 & -1 & 5 \\ 0 & 1 & f-8 \end{bmatrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 \\ 3 & & 0 \\ 4 & & \end{bmatrix}$$

$$\xrightarrow{-r_2 \to r_2} \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & -5 \\ 0 & 1 & f-8 \end{bmatrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 \\ 3 & -1 & 0 \\ 4 & & \end{bmatrix}$$

$$\xrightarrow{r_3 - r_2 \to r_3} \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & -5 \\ 0 & 0 & f-3 \end{bmatrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 \\ 3 & -1 & 0 \\ 4 & 1 & \end{bmatrix}$$

$$\xrightarrow{(1/(f-3))r_3 \to r_3} \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & -5 \\ 0 & 0 & 1 \end{bmatrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 \\ 3 & -1 & 0 \\ 4 & 1 & f-3 \end{bmatrix}$$

18. The entry in row 3 and column 3 of L is equal to:

Answer: **(b)** $f - 3$; from the previous problem, it's apparent that the entry in row 3 and column 3 of L is $f - 3$.

19. Consider the LU decomposition of the matrix $\begin{bmatrix} 2 & 1 & 2 & 0 \\ 2 & 2 & 3 & 2 \\ 4 & 3 & 6 & 3 \\ 0 & 2 & 3 & -3 \end{bmatrix}$. The entry in row 4 and column 4 of L is equal to:

Answer: **(a)** -8; using the careful book-keeping method

$$\xrightarrow{(1/2)r_1 \to r_1} \begin{bmatrix} 1 & 1/2 & 1 & 0 \\ 2 & 2 & 3 & 2 \\ 4 & 3 & 6 & 3 \\ 0 & 2 & 3 & -3 \end{bmatrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 & 0 \\ & & 0 & 0 \\ & & & 0 \\ & & & \end{bmatrix}$$

$$\xrightarrow[r_3 - 4r_1 \to r_3]{r_2 - 2r_1 \to r_2} \begin{bmatrix} 1 & 1/2 & 1 & 0 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 2 & 3 \\ 0 & 2 & 3 & -3 \end{bmatrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 2 & & 0 & 0 \\ 4 & & & 0 \\ 0 & & & \end{bmatrix}$$

$$\xrightarrow[r_4 - 2r_2 \to r_4]{r_3 - r_2 \to r_3} \begin{bmatrix} 1 & 1/2 & 1 & 0 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -7 \end{bmatrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 4 & 1 & & 0 \\ 0 & 2 & & \end{bmatrix}$$

$$\xrightarrow{r_4 - r_3 \to r_4} \begin{bmatrix} 1 & 1/2 & 1 & 0 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & -8 \end{bmatrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 4 & 1 & 1 & 0 \\ 0 & 2 & 1 & \end{bmatrix}$$

$$\xrightarrow{(-1/8)r_4 \to r_4} \begin{bmatrix} 1 & 1/2 & 1 & 0 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 4 & 1 & 1 & 0 \\ 0 & 2 & 1 & -8 \end{bmatrix}$$

then it's obvious the entry in row 4 and column 4 of L is equal to -8.

Consider the linear system $\begin{cases} x + 2y + 2z = 1 \\ 2x + 3y + 9z = -1 \\ 3x + 4y + kz = 1 \end{cases}$, $k \neq 16$.

20. The value of y is equal to:

Answer: (c) $(3k-28)/(k-16)$; putting our system into augmented matrix form

$$\left[\begin{array}{ccc|c} 1 & 2 & 2 & 1 \\ 2 & 3 & 9 & -1 \\ 3 & 4 & k & 1 \end{array}\right]$$

proceeding with Gaussian elimination

$$\xrightarrow[r_3-3r_1 \to r_3]{r_2-2r_1 \to r_2} \left[\begin{array}{ccc|c} 1 & 2 & 2 & 1 \\ 0 & -1 & 5 & -3 \\ 0 & -2 & k-6 & -2 \end{array}\right]$$

$$\xrightarrow{-r_2 \to r_2} \left[\begin{array}{ccc|c} 1 & 2 & 2 & 1 \\ 0 & 1 & -5 & 3 \\ 0 & -2 & k-6 & -2 \end{array}\right]$$

$$\xrightarrow{r_3+2r_2 \to r_3} \left[\begin{array}{ccc|c} 1 & 2 & 2 & 1 \\ 0 & 1 & -5 & 3 \\ 0 & 0 & k-16 & 4 \end{array}\right]$$

$$\xrightarrow{(1/(k-16))r_3 \to r_3} \left[\begin{array}{ccc|c} 1 & 2 & 2 & 1 \\ 0 & 1 & -5 & 3 \\ 0 & 0 & 1 & 4/(k-16) \end{array}\right]$$

then $z = 4/(k-16)$, $y = (3k-28)/(k-16)$.

21. The value of x is equal to:

Answer: (a) $(-5k+32)/(k-16)$; from the previous problem

$$x + 2\left(\frac{3k-28}{k-16}\right) + 2\left(\frac{4}{k-16}\right) = 1 \Rightarrow x = \frac{-5k+32}{k-16}$$

The matrix $A = \begin{bmatrix} 1 & 2 & 6 & 3 \\ 2 & 5 & 15 & 11 \\ 3 & 2 & 7 & -7 \\ -2 & -1 & -1 & k \end{bmatrix}$, where $k \neq 17$, is nonsingular.

22. The entry in row 1 and column 4 of A^{-1} is:

Answer: (b) $7/(k-17)$; proceeding immediately with Gauss-Jordan elimination

$$\xrightarrow[r_4+2r_1 \to r_4]{r_2-2r_1 \to r_2,\ r_3-3r_1 \to r_3} \left[\begin{array}{cccc|cccc} 1 & 2 & 6 & 3 & 1 & 0 & 0 & 0 \\ 0 & 1 & 3 & 5 & -2 & 1 & 0 & 0 \\ 0 & -4 & -11 & -16 & -3 & 0 & 1 & 0 \\ 0 & 3 & 11 & k+6 & 2 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow[r_4-3r_2 \to r_4]{r_1-2r_2 \to r_1,\ r_3+4r_2 \to r_3} \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & -7 & 5 & -2 & 0 & 0 \\ 0 & 1 & 3 & 5 & -2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 4 & -11 & 4 & 1 & 0 \\ 0 & 0 & 2 & k-9 & 8 & -3 & 0 & 1 \end{array}\right]$$

$$\xrightarrow[r_4-2r_3 \to r_4]{r_2-3r_3 \to r_2} \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & -7 & 5 & -2 & 0 & 0 \\ 0 & 1 & 0 & -7 & 31 & -11 & -3 & 0 \\ 0 & 0 & 1 & 4 & -11 & 4 & 1 & 0 \\ 0 & 0 & 0 & k-17 & 30 & -11 & -2 & 1 \end{array}\right]$$

$$\xrightarrow{(1/(k-17))r_4 \to r_4} \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & -7 & 5 & -2 & 0 & 0 \\ 0 & 1 & 0 & -7 & 31 & -11 & -3 & 0 \\ 0 & 0 & 1 & 4 & -11 & 4 & 1 & 0 \\ 0 & 0 & 0 & 1 & 30/(k-17) & -11/(k-17) & -2/(k-17) & 1/(k-17) \end{array}\right]$$

we could stop here and pay keen attention to the requirement; the entry in row 1 and column 4, which corresponds to that in the first row.

Then, zeroing out the -7 in row 1, we get the entry in row 1 and column 4 to be $7/(k-17)$.

23. The entry in row 1 and column 1 of A^{-1} is:

Answer: (d) $5(k+25)/(k-17)$; from the previous problem

$$5 + \frac{7 \times 30}{k-17} = \frac{5k - 85 + 210}{k-17} = \frac{5(k+25)}{k-17}$$